

Exercice 3

$$E = \{(x, y, z) \in \mathbb{R}^3; x + y - 2z = a\}$$

$$(0, 0, 0) \in E \text{ssi } \underline{\underline{a = 0}}$$

$$x + y - 2z = 0 \Rightarrow x = y - 2z$$

$$(x, y, z) = (y - 2z, y, z) = y(1, 1, 0) + z(-2, 0, 1)$$

$$E = \text{vect}\{(1, 1, 0), (-2, 0, 1)\} \Rightarrow \text{libre et génératrice}$$

$\Rightarrow$ Base

$$\dim(E) = 2$$